Application Integration Architecture: Pre-Built SOA

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Application Integration Architecture: Pre-Built SOA

Note: 2
Introduction 4
Challenges of soa 4
Pre-Built SOA with Application Integration Architecture 5
- Business Driven Approach 6
- Start with a Blueprint: Industry Reference Models 7
- Sustainable Integration: Process Integration Packs 8
  - Defining a Common Vocabulary: Enterprise Business Objects 9
  - Developing Consistent, Reusable Actions: Enterprise Business Services 10
  - Integrating to the Oracle Applications You Use Today 11
  - Extensibility to Meet Company-Specific Requirements – Now and Over Time 11
- Built on Best in Class Fusion Middleware 13
  - Oracle SOA Suite: Pre-Built SOA Foundation 13
  - Enterprise Service Bus (ESB): Reliable Transport Layer 13
  - Business Rules: Data-driven Business Logic 15
  - Business Activity Monitoring: Business Process Visibility 15
  - Web Services Manager: Consistent centralized security 15
  - Connectivity – Adapters: Service Enabling Your Legacy Investment 15
- Getting Off to a Strong Start: Service Oriented Architecture Governance 16
- Foundation for the Future 17

Summary: Pre built SOA 17

Application Integration Architecture: Pre-Built SOA Applications

INTRODUCTION

This white paper will discuss the challenges of SOA deployment and how these challenges can be mitigated with pre-built SOA infrastructure and applications from Oracle. It will outline how Oracle delivers Application Integration Architecture, a set of pre-built SOA applications that integrate previously isolated business functions into more seamless end-to-end processes, all powered by Oracle Fusion Middleware's SOA Suite. By using key SOA principles, companies can more quickly and easily integrate Oracle and non-Oracle applications into more valuable and flexible processes that can evolve with the business. It will also cover the technical details of the Application Integration Architecture.

CHALLENGES OF SOA

"It is apparently easier to say that a firm will adopt SOA than it is to make specific plans and follow through on them."

Randy Heffner, "Planned SOA Usage Grows Faster Than Actual SOA Usage" Forrester, March 2007

Thousands of companies are turning to Service Oriented Architecture to solve some of their biggest IT problems: integrating internal applications, reaching out to external partners, and supporting strategic business transformation. However, the path to SOA is not an easy one. In fact, industry analyst firm Forrester Research has found that while 14% of North American and European companies had planned to move to SOA in 2006, only 2% did so – demonstrating the difference between seeing the value of SOA and actually making concrete progress towards achieving it.

Figure 1: SOA Adoption from Forrester

To get the benefits of SOA, companies must move beyond point web services integration and establish a more strategic architecture, one that starts with the business process, is built on a robust architecture, and includes the methodology

and technology to manage and maintain SOA applications over time. Why is SOA so hard? Some of the barriers companies face in starting SOA initiatives include:

- Poorly understood and difficult to rationalize business processes. Without a clear business vision and well-defined processes, companies struggle to develop a top-down SOA design that can meet business needs today and over time.
- Long technology learning curves and poorly integrated components. Implementing SOA can include an overwhelming number of technologies and coding to low-level interfaces. Without the right reference architecture, business objects, and methodology, SOA work can get bogged down in unproductive details.
- Immature methodologies and governance models. Effective SOA design and reuse doesn't happen automatically. Without a way to encourage common design principles and manage services for reuse, many organizations are unable to make the organizational and process changes to effectively use SOA.

PRE-BUILT SOA WITH APPLICATION INTEGRATION ARCHITECTURE

Oracle removes some of the key inhibitors to SOA by providing both a pre-built SOA as well as industry-optimized service oriented applications, all built around a SOA governance model that encourages effective design and reuse. The result is something Oracle calls Application Integration Architecture, providing all the components of SOA in one integrated whole. Oracle delivers a SOA solution that is:

Business driven. Oracle's Application Integration Architecture begins with the key processes of an industry, and uses these processes to drive down to core SOA components such as canonical data and reference models, BPEL processes, and integration services. The result is a SOA foundation that is independent of any single application and can grow and change over time.

Packaged. Oracle provides a complete SOA software solution, including the Oracle Fusion Middleware's SOA Suite, an integrated, productive environment that is a leader in its category. Unlike other solutions, all components are designed to work together and are maintained and upgraded with new releases, reducing ongoing maintenance costs. In addition, each Oracle application provides right-sized services that integrate seamlessly into the technology stack, minimizing low-level integration work.

Manageable and maintainable. Oracle also includes key tools and methodologies for SOA governance, to help jumpstart SOA initiatives and ensure that companies continue to reap the flexibility and reuse advantages over time. These include technologies such as an extension framework and a business services repository to facilitate maintainability and reuse, as well as recommendations for setting up roles and processes for ongoing SOA management.

Business Driven Approach

"After considerable architectural and functional consideration, ScriptPro determined that Oracle's Application Integration Architecture vision provided the opportunity to leverage our Siebel CRM investment, as well as a flexible mechanism for adapting to changing business drivers."

Chris Duffy, Vice President, Information Systems & Chief Information Officer, ScriptPro

One of the key challenges of SOA is to build processes and services that truly represent the way the business works – across a wide variety of product lines, departments, and business units. Oracle brings business expertise from over 30 different industries and thousands of customer implementations to ensure that the models and processes it delivers can benefit your entire organization. By starting with the key processes of the business and using these processes – not the details of the current environment -- to drive the underlying implementation, Oracle provides solutions that deliver value whatever the source of the underlying application. In addition, Oracle Application Integration Architecture is designed to support business change, allowing users to replace components, insert new components and change business logic quickly and efficiently when needed. Oracle uses this top-down approach to deliver a complete, industry optimized solution, including:

- **Industry Reference Models.** The first task for effective SOA is to develop models based on key industry processes. Oracle Application Integration Architecture includes both the business process flows, as well as the underlying canonical models that identify the critical data to be exchanged during the process. Without the right models, each SOA element becomes yet another legacy application, with conflicting interfaces and point integrations that are difficult to adapt.
- **Process Integration Packs.** From the Industry Reference Models, specific business integration scenarios must be designed and implemented using industry standards. Oracle's run-time level business processes integrate and extend your current applications, leveraging reusable Enterprise Business Objects and Services.
- **Enterprise Business Objects and Services.** Enterprise Business Objects and Enterprise Business Services provide an application-independent representation of key business components. Think of these business objects as things like customer, product, order, etc. – the entities that describe how business operations run. Without these objects, companies often spend months developing a consolidated view of critical data across multiple systems. Enterprise Business Services are the specific actions that occur against the object; for instance, add customer, delete customer or modify customer.
- **Application Business Connector Services.** Oracle Application Integration Architecture also includes components that connect selected Oracle and partner applications to the Enterprise Business Services, providing the application-specific web services to integrate the business process with the applications you use today. Oracle will also support and maintain these integrations over time, so that Process Integration Packs continue to work as applications are upgraded – or when you move to Fusion applications.

Start with a Blueprint: Industry Reference Models

Industry Reference Models provide the blueprint for SOA Applications. By outlining key specific industry processes and activities, the models provide firms with a starting point to begin their SOA deployment. In addition, Industry Reference Models provide a method to communicate to business users about how to use and leverage service oriented concepts, by speaking to them in terms of process, not in terms of technology.

Industry Reference Models are a logical representation of the key business processes of an industry, along with a logical data model of the key information collected and used by each process. Oracle's Industry Reference Models encompass several levels of each industry process, allowing users to navigate from high-level concepts to the detailed steps of each process.

The top level (level 0) outlines an abstract high level summary of all the end-to-end processes documented for the industry. For instance, it would show Order to Cash for the high tech industry as one chevron.

From there, a drill down to level 1 models shows the steps of each process. Level 1 models show an abstract high level summary of the end-to-end flow for one of the processes outlined at level 0. For Order to Cash, it would outline the abstract steps associated with the process. For instance, it would show the steps of entering an order, fulfilling the order, and collecting cash.

Further drill down is possible to level 2 models. These models show the activities in the business process and how they are linked. At this stage, it gets quite detailed on what the steps are but is still abstract. In our example, the Order to Cash process would be outlined in more detail, such as where availability and pricing fits into the order entry process.

Finally, level 3 of the Industry Reference Models outlines specific task flows, and the detailed analytics for the task and integration points, giving a complete view of how the process would be executed.

Without these models, companies would be forced to start from scratch, wasting valuable user time defining high level concepts instead of focusing on the differences that make each company unique.

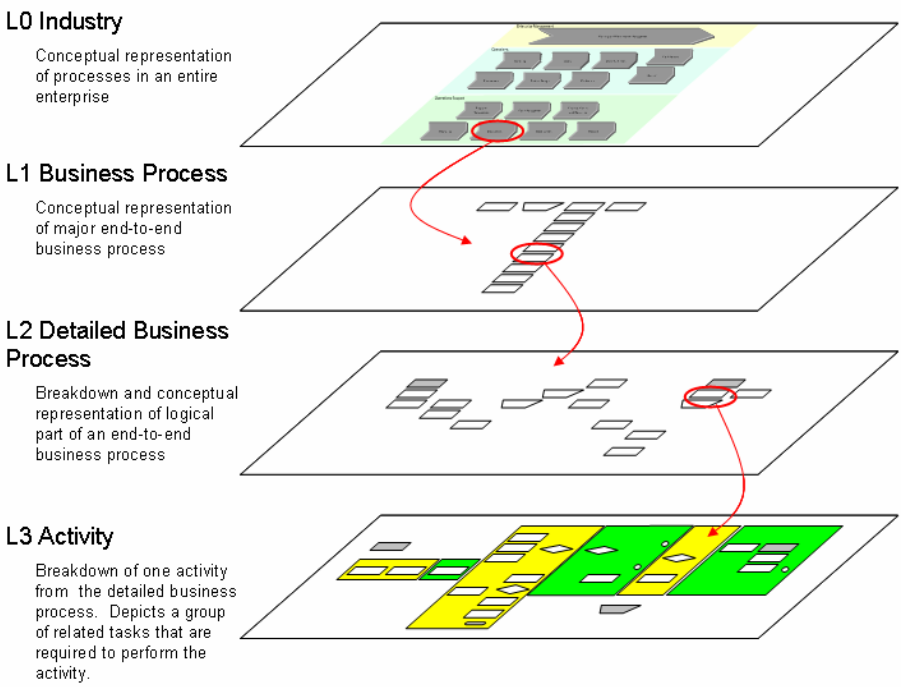

Figure 2: The levels of Industry Reference Models

Sustainable Integration: Process Integration Packs

Process Integration Packs are pre-built process integrations. They connect to the level 3 model, following the outline of what is required for each step in the Process Integration Pack. By using the Enterprise Business Objects and Services, Process Integration Packs connect Oracle applications into a higher level business process. Since they are a fully packaged product, companies can spend less time on the development process for integrations, and more time to get the integration live faster. In addition, the integrations are developed by Oracle with Oracle's full development methodology, including a robust testing process.

A good example of a Process Integration Pack would be the Order to Fulfillment integration offered between Siebel CRM and Oracle E-Business Suite Order Management. The Order Fulfillment integration provides the capability to capture an Order in Siebel CRM and then transfer it to Oracle E-Business. To start with, the product provides routines to synchronize product, customer, and pricing data between the applications. This batch mode ensures that the two applications have the same set of data to start with.

The process flow starts when a Customer Service Representative (CSR) enters an opportunity into Siebel CRM. This opportunity is captured within Siebel. As the CSR continues to refine the opportunity to include specific products, a back-end web service links to Oracle E-Business Suite Order Management to access price and availability options. That information is shown directly on the screen to the CSR, providing a seamless flow of information. Advanced configuration is also

possible by screen popping with Oracle E-Business Suite. The screens have been redesigned for a common look and feel.

After the order is completed in Siebel CRM, it is then passed to Oracle E-Business Suite Order Management using web services to transmit the completed order. To complete the transaction, Siebel CRM can inquire on the status of the order directly from the UI. This helps the CSR to provide additional insight into the status of the order by having full access to the fulfillment system.

Defining a Common Vocabulary: Enterprise Business Objects

"One of the most complicated aspects of any cross application IT project is determining the underlying canonical model. You can't communicate effectively if every application is using a different set of terms and definitions."

Erin Kinikin, Independent Analyst

Enterprise Business Objects and Services provide the foundation for Process Integration Packs, and, in fact, for SOA extensibility overall. Enterprise Business Objects are a comprehensive representation of a business object such as customer, sales order, invoice, etc. This representation contains all the attributes required for any action performed, rationalizing the differences of the underlying applications to enable a common process across them.

Enterprise Business Objects provide the shared semantic framework for applications to communicate, and are vital for Service Oriented Architectures. Semantics are the vocabulary of any service; using the same semantic framework enables various systems to understand each other. For instance, if two systems have a different concept of what a sales order is, it will be difficult for them to communicate. With a shared vocabulary, or *semantic framework*, those systems can more easily communicate. Without them, a service oriented architecture implementation increases exponentially in complication.

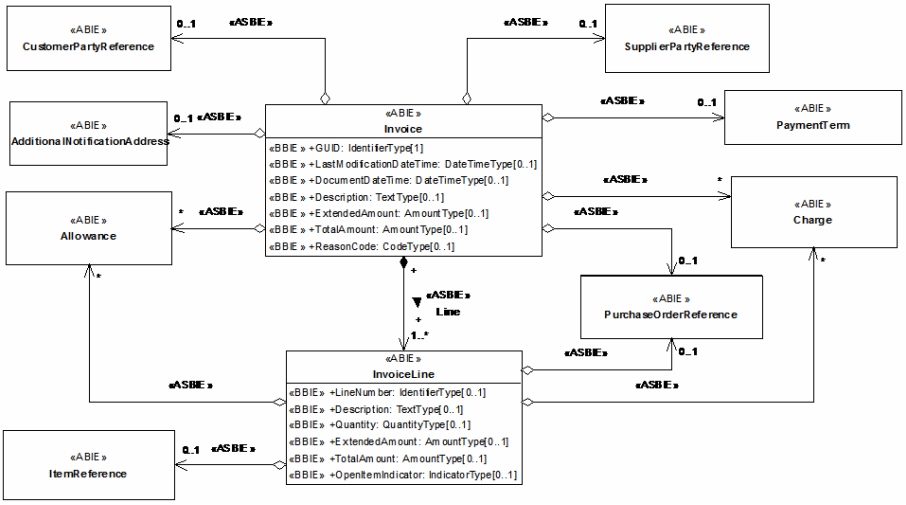

Figure 3: UML diagram for an Enterprise Business Object

Developing a common semantics framework often requires extensive discussions within the internal application groups trying to come up with a fair representation of what the object is. Oracle has done this work for you, using the best of standards from the Open Application Group, as well as the experiences of tens of thousands

of customer implementations across its wide set of product lines. With Oracle Enterprise Business Objects, time can be focused on what makes the object unique in a specific environment, not what the base definition is. In many cases, firms revert to point to point integration over a more robust method since a semantic framework cannot be agreed on. This minimizes the reuse value associated with Service Oriented Architecture.

Enterprise Business Objects play two important roles in SOA implementations: first, they provide a mechanism to plug in different applications to an existing business process. Once any application maps to the common format of the relevant Enterprise Business Objects, it can participate in Process Integration Packs. This means that companies can map in their own legacy system functionality or another packaged application when required, and still get all the benefits of the Oracle industry processes, models, and the other integration points. Of course, inserting a new application will require additional testing, but with the framework of Enterprise Business Objects, substituting applications can easily be achieved.

Secondly, Enterprise Business Objects provide a central semantic framework that all services can map to. Since one only needs to map to the common definition, not all possible participating applications, the number of connections is reduced, decreasing development time and reducing maintenance. This enables a reduction in the time IT spends on maintenance, freeing them to spend more time on new business projects.

Developing Consistent, Reusable Actions: Enterprise Business Services

Enterprise Business Services are closely linked to Enterprise Business Objects. The Enterprise Business Services provide the web service implementation of the Enterprise Business Object. During the development process of Enterprise Business Objects, the key actions that need to be performed on the object are determined. These actions are generic things such as "add," "modify" or "delete." They can also be specific to the object, such as "get item availability." These actions are realized as Enterprise Business Services.

Like the Enterprise Business Objects, the Enterprise Business Services are not tied to any implementation. Enterprise Business Services can be bound to various back-end systems, but they are only loosely coupled to them. This enables multiple systems to offer the same service and facilitates reuse.

This is especially helpful when multiple applications perform similar business functions within the IT landscape. Oracle Business Rules provides a mediating layer to route a service request to the right back-end system based on the content of the request. This allows companies to more easily adjust to changing business conditions – such as the addition of a new supplier – without complex coding. Using Oracle Business Rules, companies can even choose to implement a service with multiple back-ends, using rules to determine when each specific back-end system is most appropriate.

For instance, in the telecommunications industry, industry consolidation and diverse product lines have resulted in multiple systems performing common tasks such as billing. Depending on the contents of the request, a different system may need to provide the balance. So when an inquiry is made about a bill, the right fulfillment system can be used without the requestor knowing which system fulfills that request, or even that the fulfillment mechanism has changed. The abstraction layer provided by Enterprise Business Services allows for one call to be executed against multiple systems.

Integrating to the Oracle Applications You Use Today

One of the most tedious parts of SOA is to service-enable each participating application. Oracle is approaching this problem from two angles: First, selected Oracle applications will deliver right-sized native web services that make them easier to integrate with the Process Integration Packs process flows. These web services are based on the business activities to be supported rather than by the legacy APIs of each application. These right-sized web services will make for a more sustainable service oriented architecture. Instead of having to sequence together granular API calls, the right-sized web service provides a single, more usable interface. This makes for less work when applications are upgraded and when interfaces change.

These right-sized services are not limited to use in Process Integration Packs. The services for each application are available to all customers, and can be used for custom developed integrations as well. The services can also be extended to add business-specific logic or other processing. These services will be discoverable through the use of the Business Services Repository, which is covered later in more detail.

The second aspect of service enablement Oracle addresses is connecting the web services of a specific application to the Enterprise Business Objects and Services. This is done using something called an Application Business Connector service. ABCs provide a kind of "bridge" between the common model and the specifics of a given underlying application. For example, in the case of the telecommunications inquiry, the ABC service for each fulfillment service could validate and map inputs, using cross reference keys to translate to application specific values like customer id. So when an incoming request for getting a customer's current balance Enterprise Business Service is invoked, the business rules are executed and routed to the correct Application Business Connector service, which maps to the back-end system.

Extensibility to Meet Company-Specific Requirements – Now and Over Time

One of the keys to achieving the benefits of SOA is to design services that can be modified, extended, and maintained quickly and efficiently. Oracle recognizes that not all companies are the same, and processes may have to be altered – both

initially and over time as the business changes. In Application Integration Architecture, the alteration occurs through extension wherever possible.

It is vital to understand what exactly extensibility is. Most software products in the market place will allow you to modify the delivered components to meet custom requirements. This often involves duplicating the object and recoding it, which while effective in altering the process, takes significant cycles of development and testing, not to mention the additional effort during maintenance. Oracle has designed its objects to facilitate extension instead of coding to change the behavior.

Extensions are unique in two ways:

1. By providing well-defined mechanisms to extend EBOs and EBSs, Oracle makes these objects easier to support and maintain. For example, Oracle provides a methodology and tools to add fields to an Enterprise Business Object. Business Services have well-defined pre- and post- processing exits where additional processing can be performed. Once these extensions are added, when an upgrade is performed on the integration, the new elements can be carried forward to the new version.

2. Extensions are easier to trace, audit, and control. For instance, with Enterprise Business Objects, all extensions are kept in a separate file than with the base Enterprise Business Object definition. This separation makes it easier to audit when extensions are performed, it also makes it easier to trace and debug integrations since the extensions are isolated.

Oracle intends to use an extension paradigm, over a customization one, whenever possible in the development of Oracle Application Integration Architecture. Extensions also apply not only to application behavior, but also to data elements. Enterprise Business Objects, and in turn Enterprise Business Services, can be extended to meet specific data requirements.

This means that not only will changing the behavior of Oracle Application Integration Architecture be easier than other solutions (due to reduced coding) but also that extensions can be preserved during an upgrade.

Built on Best in Class Fusion Middleware

Fusion Middleware ranked as Leader in the following Gartner Magic Quadrants

- Application Platform Suites
- Development Tools
- Application Server
- Web Services Platform
- Enterprise Portal
- Business Integration
- Identity Management
- Web Services Management
- ETL Data Integration

Until now, this paper has been focused on the content aspects of Service Oriented Architecture and Application Integration Architecture, but what about the technology? To truly have a pre-built SOA, the underlying technology must be solid. Oracle has just that in Oracle Fusion Middleware.

Oracle Fusion Middleware is the world's fastest growing family of middleware solutions, spanning from SOA, portals, and process management, to application infrastructure, identity management, content management, and business intelligence. By buying all these elements from a single vendor, companies get a more integrated, supportable solution.

Oracle SOA Suite: Pre-Built SOA Foundation

Oracle SOA Suite is a comprehensive software suite for the building, deployment, and management of a Service-Oriented Architecture. This includes the service-oriented development of applications, service-oriented integration of applications and IT systems, and business process management (BPM). The components of the suite benefit from common capabilities including a single deployment and management model, consistent tooling, end-to-end security, and unified metadata management.

"What's exciting about the new Oracle SOA Suite 10g Release 3 is the comprehensive scope and breadth of its well-integrated component set that's geared to provide just such an environment."

- James Borck, "Oracle Sows the Seeds for SOA," Infoworld Jan 2007

Oracle SOA Suite consists of (i) a multi-protocol Enterprise Service Bus (ESB) to connect applications and route messages; (ii) a BPEL-based Process Manager to compose services into business processes; (iii) a business rules engine to capture and automate business policies; (iv) a Business Activity Monitoring (BAM) solution to gain real-time visibility into the operation and performance of business processes and services; (v) a Web services management and security solution to enforce authentication and authorization policies on services; and (vi) connectivity to virtually any data source including applications, databases, queues, RFID and other physical devices; and high volume high performance data integration.

Enterprise Service Bus (ESB): Reliable Transport Layer

Oracle ESB is a high-performance multi-protocol bus that provides messaging, routing, and transformation capabilities to easily integrate applications, services, and IT systems. Application Integration Architecture, (specifically the Process Integration Packs) relies heavily on the Oracle ESB. The ESB provides the layer that can expose the Enterprise Business Services and related Application Business Connector services. Applications are connected to the ESB through either native web services support of the application or through an adapter. Once connected, the Application Business Connector services run in the ESB– taking native application objects and transforming them to Enterprise Business Objects for outbound requests and the reverse for inbound requests. The Enterprise Business Services are coupled to the Application Business Connector services, also running in the ESB.

Oracle ESB provides connectivity leveraging Oracle Adapters, which provide standards based access to virtually any data source or legacy application. This

means that even applications that may not be completely web service enabled can participate in Application Integration Architecture.

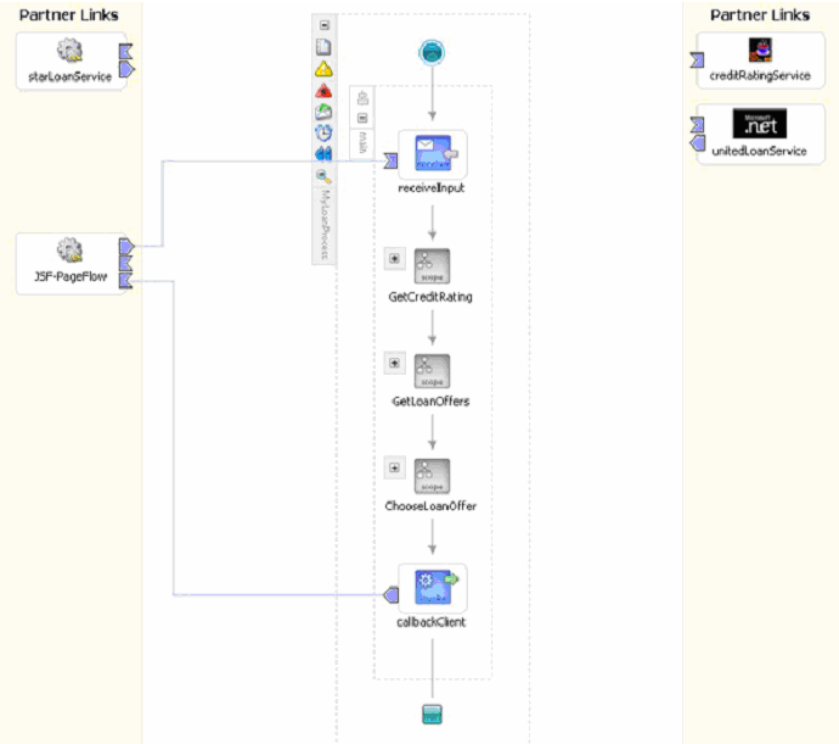

Figure 4: Graphical Development of Business Processes in BPEL

BPEL Process Manager: Graphical Process Manager

"BPEL will emerge as the leading industry standard for Web services flow composition (0.8 probability)."

—David Smith
Gartner

Oracle BPEL Process Manager provides a comprehensive, standards-based and easy to use solution for creating, deploying, and managing cross-application business processes with both automated and human workflow steps. Oracle BPEL Process Manager provides the orchestration aspects for Application Integration Architecture. When additional logic is required to sequence services together exposed through the Oracle ESB, Oracle BPEL Process Manager acts as the guide to connect those services into a process.

Oracle BPEL engine is the most mature, scalable, and robust BPEL server available today. It executes standard BPEL processes and provides a "dehydration" capability so that the state of long-running flows is automatically maintained in a database, ensuring that business processes can scale across multiple systems and that the processes can continue with no loss of process status or underlying data if an underlying system is disrupted.

Oracle BPEL Process Manager features a graphical and user-friendly Process Designer, a Web-based Console for management, administration, and debugging of deployed processes, making it easier for developers to modify and extend Process Integration Packs.

Business Rules: Data-driven Business Logic

Oracle Business Rules is a fast, efficient, and standards based rules product that enables business analysts to easily define, update, and manage key decisions and policies governing business processes and applications, for example, business policies within business processes that are likely to change can be captured using business rules.

Oracle Business Rules can be used throughout Application Integration Architecture to ensure that the right service is always used for each process step, and to add business logic and routing rules without coding.

Business Activity Monitoring: Business Process Visibility

Oracle Business Activity Monitoring (BAM) is a complete solution for building real-time operational dashboards to monitor business processes and services, services levels, and to track key performance indicators (KPIs) from processes and services, with capabilities to take automatic or manually invoked corrective actions.

"BAM defines the concept of providing real-time access to critical business performance indicators to improve the speed and effectiveness of business operations"

—Bill Gassman, Analyst
Gartner Group

Oracle BAM enables business users to build interactive, real time dashboards, and proactive alerts. It leverages the latest web technology to deliver a rich, interactive operational dashboard in which real time data and personalized reports are delivered to business users via a standard web browser. Oracle BAM is built on a unique high-performance, event-driven, memory-resident architecture specifically designed for the needs of real-time analytics and reporting applications.

With the connectivity associated with SOA Suite between components, it's easy to build Oracle BAM dashboards leveraging processes built in Oracle ESB and Oracle BPEL Process Manager.

Web Services Manager: Consistent centralized security

Oracle Web Services Manager (OWSM) is a comprehensive solution to secure and manage service-oriented architectures. OWSM allows administrators to centrally define policies that govern web services within Application Integration Architecture such as authentication and access, logging, and content validation, without requiring any modification to the existing web services.

Administrators can set the quality of service levels for each service and configure parameters for alerts when an application exceeds established targets. A graphical monitoring dashboard provides IT operations staff with real-time visibility into the health, performance, security, and utilization of web services. This enables for integrations developed within Application Integration Architecture to be secured, and monitored to ensure that they meet an established level of service.

Connectivity – Adapters: Service Enabling Your Legacy Investment

Oracle Adapters enable connectivity to virtually any data source inside the enterprise. Oracle Adapters are available for more than 300 packaged applications, along with legacy and mainframes systems including Tuxedo, VSAM, CICS, and

other technologies including FTP, Files, Database, AQ, and JMS. Oracle Adapters are standards-based and support both web services and JCA.

Getting Off to a Strong Start: Service Oriented Architecture Governance

The promise of SOA is clear: to free up your application assets to work in new ways. However, without a way to manage these assets, it is often difficult to ensure that they are used (and reused) appropriately. In order ensure that Oracle Application Integration Architecture can be managed and maintained as part of a larger SOA initiative, Oracle provides the following tools and methodology components for effective SOA governance:

- **Extension Methodology.** Oracle intends to provide a methodology, along with the extension framework discussed above, that allows companies to modify and extend the solutions and still participate in Oracle support and upgrades over time.
- **Suggested Roles.** Oracle intends to provide documentation on what roles are performed and the best practices for each role. For instance, one role suggested by Oracle is Integration Administrator. This role is key to mediating disputes during integration testing and deployment. Since multiple teams are often involved within an integration project, it can be challenging to determine where and how errors are occurring. By having a centralized role to isolate and determine which group takes the lead on addressing potential issues, integrated testing is completed faster and production problems are resolved more quickly.
- **Reference Architecture.** Oracle intends to provide materials that describe how to use Application Integration Architecture. The reference architecture outlines the key components, and their role in robust integration development, providing a framework for other integration projects.
- **Design Patterns.** Oracle intends to provide documentation that outlines how integrations and processes should be developed, along with how to invoke and use services. This makes development more efficient by giving IT organizations a starting point for development. Also, when used consistently, maintenance is easier.
- **Test Harness.** A test harness is intended for delivery that enables integration tests to be automatically configured and run. By offering an end-to-end test harness, Oracle provides a way for firms to ensure that any changes or custom integrations are safe to deploy.
- **Business Services Repository.** The Business Services Repository will provide a simple, intuitive way to discover, deploy, track, and manage the services that make up Application Integration Architecture. Since all the services will be registered in Business Services Repository, developers can use an intuitive web UI to discover services for use in other integrations.

These are just a few of the elements of SOA governance provided with Application Integration Architecture. Oracle provides additional SOA governance components with Oracle Fusion Middleware's SOA Suitew.

Foundation for the Future

Not only does Application Integration Architecture offer a platform for customers to derive value today, it also offers them a path forward. Application Integration Architecture is a product customers can adopt to get to Fusion Applications:

1. **Processes and Models.** The models and industry processes will serve as the foundation for Fusion Applications.
2. **Architecture.** The Application Integration Architecture uses key elements that will make up Fusion Applications. Investments in this architecture will continue to offer value with Fusion Applications.
3. **Fusion Applications Application Compatible.** With the release of Fusion Applications, Oracle intends to make sure that Fusion Application modules can be plugged into Application Integration Architecture, enabling customers to upgrade to Fusion and retain their investments. Customers can be assured that investments in Application Integration Architecture will continue to have value even after they adopt Fusion Applications.

SUMMARY: PRE BUILT SOA

Oracle Application Integration Architecture offers an effective starting point for companies embarking on Service Oriented Architecture initiatives. It includes pre-built business processes and the service oriented architecture to deploy quickly and manage these processes effectively over time. Using Oracle Application Integration Architecture, companies can achieve:

- **Rapid business value**. Oracle's Application Integration Architecture is designed to solve the key integration issues for a company's most strategic business processes. By starting with pre-built industry models, companies can minimize up-front discovery time and focus only on the elements that are unique to their business.
- **Lower integration and maintenance costs.** Oracle provides a complete SOA software solution, including the Oracle Fusion Middleware stack, an integrated, productive environment that is a leader in its category. Unlike other solutions, all components are designed to work together and are maintained and upgraded with new releases, reducing ongoing maintenance costs.
- **Manageable and maintainable SOA applications.** Oracle also includes key tools and methodologies for SOA governance, to help jumpstart SOA initiatives and ensure that companies continue to reap the flexibility and reuse advantages over time.

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